IT4010 Research Project – 2025 July

Topic Assessment Form

Project ID : 25-26J-202

1. Topic (12 words max)

Smart Agriculture Advisory Platform for Coconut Plantations in Sri Lanka

2. Research group the project belongs to

SST - Software Systems & Technologies

3. Specialization of the project belongs to

Information Technology (IT)

4. If a continuation of a previous project:

Project ID	
Year	

5. Brief description of the research problem including references (200 – 500 words max) – references not included in word count.

Coconut cultivation is one of the most vital agricultural sectors in Sri Lanka, contributing significantly to rural livelihoods, food security, and national exports. However, the industry faces growing challenges due to inefficient farming practices, climate change, pest outbreaks, and lack of access to modern agricultural intelligence. Many smallholder coconut farmers still rely on manual decision-making and fixed schedules for irrigation and harvesting, leading to water wastage, unpredictable yields, and crop health deterioration.

With rising climate variability, traditional farming approaches are increasingly unreliable. Irregular rainfall and prolonged dry periods make it difficult to maintain optimal soil moisture levels. Farmers often either over-irrigate or under-irrigate, negatively affecting productivity. Additionally, coconut yield estimation is rarely data-driven, making it hard for farmers and stakeholders to plan for storage, marketing, or export cycles. Furthermore, pests and diseases such as red palm weevil, bud rot, and leaf spot are often detected too late, resulting in irreversible damage to trees.

Another overlooked aspect is the overall health status of a coconut plantation. Farmers lack a unified scoring system to monitor and assess the plantation's vitality over time. Without insights into factors like soil condition, irrigation sufficiency, disease presence, and climate impacts, it is difficult to make holistic decisions about farm management and long-term sustainability.

This research proposes a Smart Agriculture Advisory Platform for Coconut Plantations in Sri Lanka, integrating four intelligent modules:

1. Irrigation Recommendation Engine – uses machine learning to optimize water usage based on soil moisture and weather forecasts.
2. Crop Advisory & Yield Prediction Engine – provides data-driven coconut suitability checks and yield forecasts based on agro-environmental factors.
3. Disease Detection & Farmer Notification System – leverages image processing and classification models to detect diseases early and notify farmers in local languages.
4. Plantation Health Scoring System – combines multisource data to generate a visual, interpretable score representing the overall health of the plantation.

By using localized datasets and farmer-friendly interfaces in Sinhala and Tamil, this platform aims to promote precision agriculture, resource efficiency, and climate resilience in Sri Lanka's coconut sector.

References

[1] R. M. Pathiraja, G. Vitharana, and D. Jayasinghe, "Current status and future challenges in coconut cultivation in Sri Lanka," *Journal of the Coconut Research Institute of Sri Lanka*, vol. 27, no. 1, pp. 45–52, 2020.

[2] M. S. Uddin, S. H. Baten, and S. Debnath, "Smart irrigation system using machine learning and IoT," in *Proc. 5th Int. Conf. on Electrical Information and Communication Technology (EICT)*, Dhaka, Bangladesh, 2019, pp. 1–6.

[3] F. S. Prity, M. R. Islam, and A. Ahmed, "Machine learning-based crop recommendation system using real-time sensor data," *Human-centric Intelligent Systems*, vol. 3, no. 4, pp. 497–510, 2022.

[4] R. Alkhawaji, S. M. Rehman, and M. Alahmadi, "Coconut yield prediction using Bi-LSTM with optimized training algorithms," *Applied Sciences*, vol. 14, no. 17, pp. 1–16, 2024.

[5] P. H. D. Priyadarshana and H. M. T. D. Herath, "Early detection of coconut leaf diseases using deep learning," in *Proc. Int. Conf. on AI and Agriculture (AIA)*, Colombo, Sri Lanka, 2023, pp. 121–126.

[6] N. H. Jayasinghe, S. Weerasinghe, and M. Dissanayake, "Towards a smart agriculture framework for Sri Lankan smallholder farmers," *ICT for Agriculture Journal*, vol. 2, no. 1, pp. 10–20, 2022.

6. Brief description of the nature of the solution including a conceptual diagram (250 words max)

The proposed solution is a **Smart Agriculture Advisory Platform** designed specifically for coconut plantations in Sri Lanka. It integrates machine learning models and cloud-based APIs to deliver intelligent, real-time decision support to farmers via a web and mobile dashboard.

The system architecture comprises four key components:

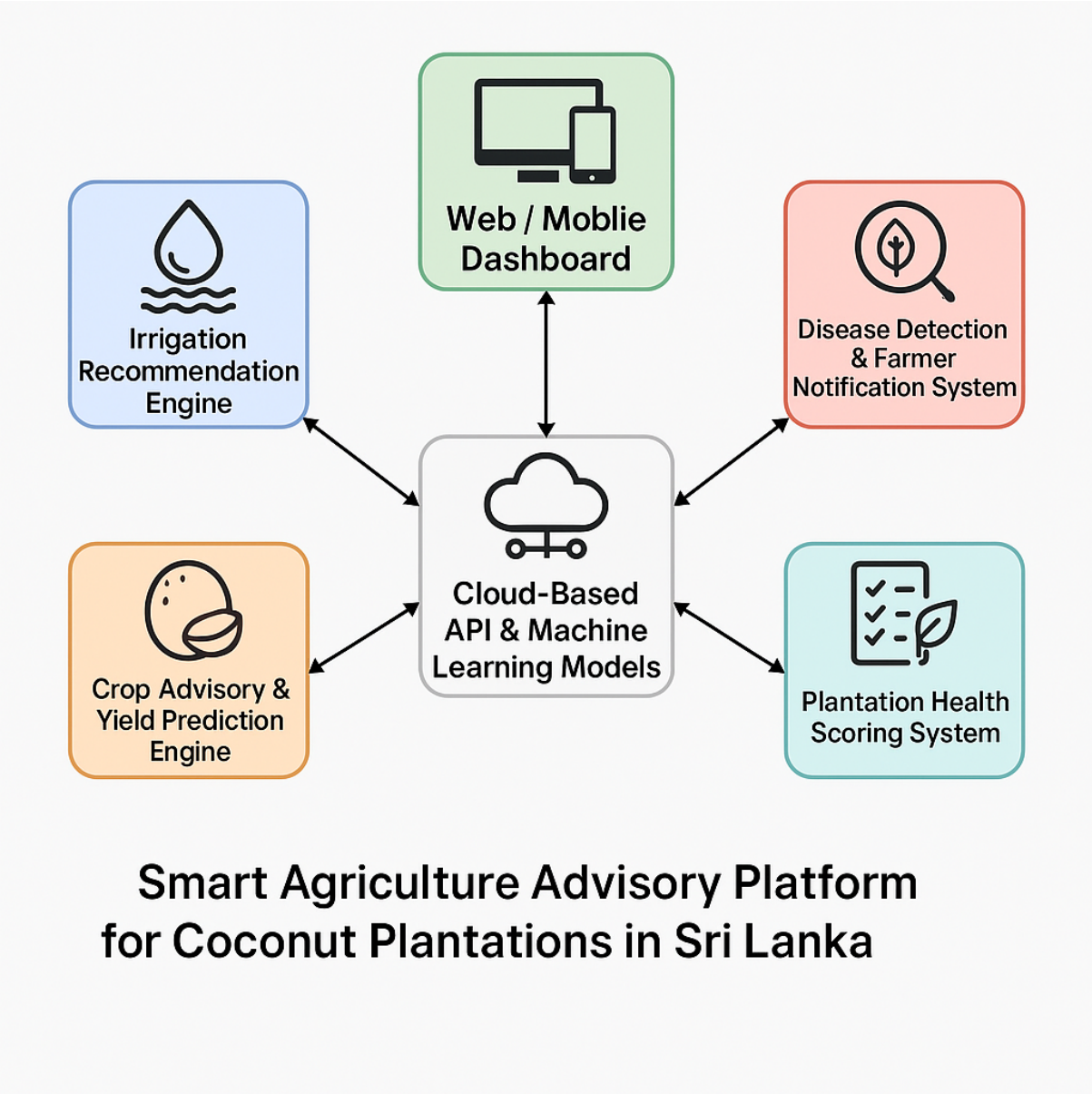
1. **Irrigation Recommendation Engine**: This module analyzes soil moisture history, rainfall forecasts, and weather parameters using a classification model (e.g., Random Forest) to recommend optimal irrigation schedules. The output is tailored to regional agro-climatic patterns to reduce water waste and promote sustainability.

2. **Crop Advisory & Yield Prediction Engine**: Focused solely on coconut cultivation, this component evaluates environmental conditions such as soil pH, NPK levels, and weather data to determine suitability for coconut and predict expected yield using models like Bi-LSTM or XGBoost.

3. **Disease Detection & Farmer Notification System**: This module employs image-based classification (e.g., CNN) to detect early symptoms of common coconut diseases and sends multilingual alerts to farmers through SMS or dashboard notifications.

4. **Plantation Health Scoring System**: Aggregates inputs from all modules—irrigation status, disease presence, soil condition, and yield projections—to generate a visual score that reflects the overall health of the plantation.

All modules interact via secure RESTful APIs, and predictions are visualized on a user-friendly, multilingual dashboard. The system empowers farmers with timely insights to increase productivity, reduce losses, and make data-driven decisions.


Web / Moblie Dashboard
Irrigation Recommendation Engine
Disease Detection & Farmer Notification System
Cloud-Based API & Machine Learning Models
Crop Advisory & Yield Prediction Engine
Plantation Health Scoring System
Smart Agriculture Advisory Platform for Coconut Plantations in Sri Lanka

7. Brief description of specialized domain expertise, knowledge, and data requirements (300 words max)

The development of the **Smart Agriculture Advisory Platform for Coconut Plantations in Sri Lanka** demands interdisciplinary knowledge and domain-specific data to ensure the platform's accuracy, usability, and long-term value to farmers.

Domain Expertise

- **Agronomy & Coconut Cultivation**
  Expertise in coconut crop cycles, nutrient needs, water sensitivity, and pest/disease behavior is essential. Collaboration with agricultural experts and institutions (e.g., Coconut Research Institute) is key.
- **Machine Learning & Artificial Intelligence**
  Knowledge in supervised learning models (e.g., Random Forests, Decision Trees), deep learning (e.g., LSTM, CNNs), and time-series prediction is required for building irrigation, yield, and disease advisory modules.
- **Software Engineering & Cloud Development**
  Skills in API development, cloud hosting, and frontend frameworks are needed to deploy scalable systems and build a farmer-friendly, multilingual dashboard.

Knowledge Requirements

- Agro-ecological zoning and climate behavior in Sri Lanka
- Soil chemistry and interpretation of pH and NPK data
- Identification and classification of coconut diseases (leaf rot, bud rot, etc.)
- Seasonal patterns and irrigation practices in coconut farming

Data Requirements

- **Environmental Data**: Localized temperature, rainfall, humidity (e.g., from Dept. of Meteorology, NASA POWER).
- **Soil & Crop Data**: pH, NPK test results, plantation-level soil maps.
- **Yield Records**: Historical coconut yield reports from Coconut Development Authority.
- **Disease Image Dataset**: Annotated images of infected coconut plants (from field visits or agricultural databases).
- **Farmer Input Logs**: Irrigation history, fertilizer schedules, pest incidents for model training and validation.

This blend of expertise and data ensures the system is grounded in scientific precision while being practical for real-world deployment.

8. Objectives and Novelty

Main Objective - To develop a Smart Agriculture Advisory Platform for Coconut Plantations in Sri Lanka using machine learning to support irrigation scheduling, yield forecasting, disease detection, and overall plantation health assessment.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Prasangi W.K.K.I IT21215674	Develop an Irrigation Recommendation Engine	• Collect historical soil moisture, temperature, and rainfall data from coconut-growing regions • Preprocess and normalize datasets • Train and evaluate machine learning models (Random Forest / SVM) • Develop an API endpoint to provide irrigation predictions • Integrate the model with frontend dashboard for visual recommendations • Test with varying climate zone inputs	A region-specific machine learning engine that dynamically recommends coconut irrigation volumes using localized Sri Lankan climate and soil data

Niwantha W.P.I IT21239366	Build Crop Advisory & Yield Prediction Engine (Coconut-specific)	• Analyze agro-environmental data including pH, NPK, temperature, and rainfall • Train a classification model to assess coconut suitability • Collect and analyze historical coconut yield records • Train Bi-LSTM/XGBoost models for yield prediction • Create intuitive visualizations for predicted yield and crop advice • Integrate the logic into the main dashboard	First platform offering coconut-specific suitability advice and long-term yield forecasting tailored to Sri Lankan agro-climatic regions
Fernando W.T.T.S IT22590244	Implement Disease Detection & Farmer Notification System	• Gather a dataset of coconut leaf and trunk disease images • Label and augment image data for training • Train a CNN model to detect diseases like leaf spot, bud rot, etc. • Build an alerting system (email/SMS/UI) • Implement multilingual notifications (Sinhala/Tamil) • Integrate detection module with the farmer dashboard	Mobile-first, image-based early disease detection system for coconut plantations with localized language support for real-time farmer alerts
SANKALPA T.W.K.S.D IT22564290	Develop a Plantation Health Scoring System	Define scoring parameters from all modules (irrigation, yield, disease, climate, etc.) • Assign weightings to different indicators (e.g., moisture level, disease	Integrated plantation health scoring mechanism for coconut farming that combines ML outputs to give a holistic and interpretable overview of farm performance

		probability) • Develop a scoring algorithm to compute health score per plot • Design dashboard visuals to show color-coded plantation health over time • Provide recommendations based on score trends	

9. Individual component description of how it is complied with the specialization.

Member Name with Registration No	Description
Prasangi W.K.K.I IT21215674	The member is responsible for developing the **Irrigation Recommendation Engine**, which applies supervised machine learning techniques such as Random Forest or SVM to generate optimal irrigation schedules based on soil moisture and weather data. This work involves data preprocessing, model training, API development, and integration with the user interface—demonstrating key IT skills in data science, backend development, and software integration.
Niwantha W.P.I IT21239366	This member handles the **Crop Advisory & Yield Prediction Engine (Coconut-specific)**. The module includes training models like Bi-LSTM or XGBoost to forecast coconut yield and assess the suitability of environmental conditions for coconut cultivation. The task showcases IT expertise in AI model development, data analytics, predictive systems, and full-stack application development.
Fernando W.T.T.S IT22590244	This member is responsible for the **Disease Detection & Farmer Notification System**, which involves training a CNN-based image classification model to identify coconut diseases at early stages. The system includes real-time detection and multilingual alert delivery. It incorporates image processing, model deployment, cloud services, and UI/UX—key areas within IT and intelligent system design.
SANKALPA T.W.K.S.D IT22564290	The member is assigned to build the **Plantation Health Scoring System**, which aggregates inputs from other modules and calculates an overall health score using weighted algorithms. The score is visualized on the dashboard using charts and color indicators. This work leverages core IT skills in algorithm design, data modeling, backend API logic, and information visualization.

10. Supervisor details

	Title	First Name	Last Name	Signature
Supervisor	Mr	Uditha	Dharmakeerthi	
Co-Supervisor	Ms	Dinithi	Pandithage	
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

<u>This part is to be filled by the Topic Screening Staff members.</u>

a) Does the chosen research topic possess a comprehensive scope suitable for a final-year project?

Yes		No	

b) Does the proposed topic exhibit novelty?

Yes		No	

c) Do you believe they have the capability to successfully execute the proposed project?

Yes		No	

d) Do the proposed sub-objectives reflect the students' areas of specialization?

Yes		No	

e) Supervisor's Evaluation and Recommendation for the Research topic:

Acceptable: Mark/Select as necessary

Topic Assessment Accepted	
Topic Assessment Accepted with minor changes*	
Topic Assessment to be Resubmitted with major changes*	

Topic Assessment Rejected. Topic must be changed	

* Detailed comments given below

Comments

Staff Member's Name	Signature

*__Important__:

1. According to the comments given by the evaluator, make the necessary modifications and get the approval by the **Evaluator**.
2. If the project topic is rejected, identify a new topic, and request the RP Team for a new topic assessment.